

UNIT III — FUNGI

Definition — Fungi: Eukaryotic, non-photosynthetic organisms that obtain nutrients by absorption. Most have a chitin-rich cell wall and commonly reproduce by spores.

3.1 General characteristics

- Eukaryotic organization with true nuclei.
- Lack chlorophyll and do not perform photosynthesis.
- Nutrition is absorptive: enzymes digest material externally and soluble products are absorbed.
- Reserve food commonly includes glycogen and lipids.
- Body may be unicellular or filamentous.
- Filamentous body consists of hyphae; a mass of hyphae is mycelium.
- Reproduction may be vegetative, asexual and sexual.

3.2 Thallus organization

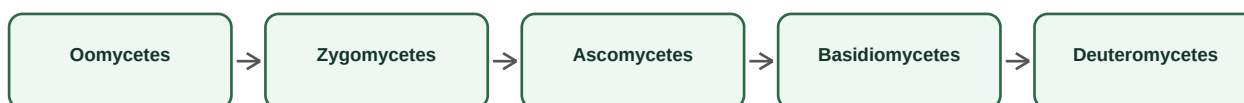
Definition — Thallus: A plant-like body not differentiated into true root, stem and leaf.

Type	Description
Unicellular	Single-celled fungal body; common in yeasts
Coenocytic / aseptate hyphae	Continuous multinucleate hyphae with few or no cross walls in the classical sense
Septate hyphae	Hyphae divided by septa into compartments, often with pores
Pseudomycelium	Chain-like arrangement of elongated budding cells in some yeasts

3.3 Classification — Ainsworth (1973) framework

The syllabus specifically names **Ainsworth (1973)**. Historical classification systems remain important for examinations even though modern phylogenetic classification has changed relationships.

Exam-oriented major groups named in the syllabus



Exam focus: Write “according to the Ainsworth (1973) syllabus framework” and do not mix the historical exam classification with a modern DNA-based classification unless asked to compare them.

3.4 Modes of reproduction

Vegetative reproduction

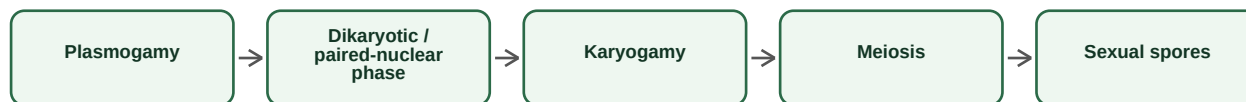
- Fragmentation of mycelium.
- Budding in suitable unicellular forms.
- Other vegetative propagules in selected fungi.

Asexual reproduction

Asexual reproduction commonly produces mitospores. Classical categories include zoospores, sporangiospores and conidia, depending on the group.

Sexual reproduction — general sequence

General sexual sequence in fungi



Definition — Plasmogamy: Fusion of compatible cytoplasm.

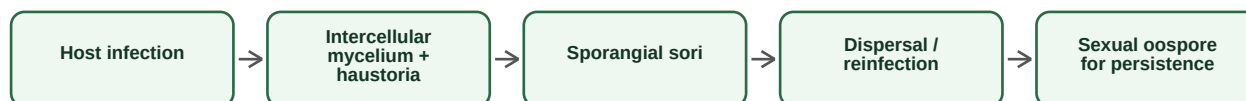
Definition — Karyogamy: Fusion of compatible nuclei.

3.5 Major groups and representative genera

Oomycetes — *Albugo*

Albugo is classically studied as a representative causing white rust disease in crucifers. Key points: intercellular parasitic mycelium, haustoria, sporangial sori and sexual reproduction producing an oospore in the classical life-history account.

Simplified *Albugo* life-history logic



Zygomycetes — *Mucor*

Mucor is a classical example of a coenocytic saprophytic fungus. Asexual reproduction occurs through sporangia and sporangiospores; sexual reproduction between compatible strains leads to a resistant zygospore.

Ascomycetes — *Peziza*

Peziza illustrates an ascocarp, often cup-shaped (apothecium), containing asci. Sexual spores, called ascospores, are produced inside asci.

Basidiomycetes — *Puccinia*

Puccinia is a rust fungus important in plant pathology. Its life cycle is classically taught through specialized spore stages and, in heteroecious species, association with different hosts. Follow the specific spore-cycle diagram prescribed by your teacher.

Deuteromycetes — *Alternaria* and *Cercospora*

These are classically grouped by the absence of a known sexual stage in the older classification framework. They commonly reproduce by conidia. **Alternaria** is known for dark, often multicellular conidia, while **Cercospora** includes important leaf-spot pathogens.

Group	Representative	Key exam feature
Oomycetes	<i>Albugo</i>	White rust; oospore in classical account
Zygomycetes	<i>Mucor</i>	Coenocytic mycelium; sporangia; zygospore
Ascomycetes	<i>Peziza</i>	Asci with ascospores; apothecium
Basidiomycetes	<i>Puccinia</i>	Rust fungus with complex spore stages
Deuteromycetes	<i>Alternaria</i> / <i>Cercospora</i>	Conidial reproduction emphasized in classical system

3.6 Economic importance of fungi

Beneficial	Harmful
Decomposition, edible mushrooms, fermentation, industrial enzymes, antibiotics, other metabolites and mycorrhizal associations	Plant diseases, food spoilage, mycotoxins, human/animal diseases and material deterioration

Exam focus: Unit III must-master diagrams: generalized hypha, Mucor sporangium, Peziza ascus/apothecium, Puccinia concept chart and Alternaria conidium.